Effect of CuO on laser absorption in glass to glass laser bonding

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***Abstract*—Localized laser bonding using a glass frit intermediate layer is an ideal technology to hermetically package miniature devices without heating the function components. It is important to describe the laser energy transmission and absorption in glass frit when modeling heat transfer in transmission laser bonding of glass to glass. Excess energy leads to over-burning of the glass material and deficit energy produces weak bonding. In this paper, the effect of CuO as additive in glass frit is investigated. The research shows that the content of CuO is confirmed to have direct influence on the laser absorption and the bonding quality. Shear force test, SEM and EDX analysis are used to survey the bonding quality. After a series of experiment and analysis, we found that adding 15% CuO can get a best result of encapsulation. Within the scope of this add, the glass frit could meet the needs of packaging.**

Keywords—TLB; glass to glass laser bonding; absorptivity of laser

I. INTRODUCTION

With the improvement of electronic integration, the size of encapsulation is developing toward to the submicron and even deep sub-micron. Many miniature optical, electronic, and micromechanical systems, for example microelectromech-anical systems (MEMS), require hermetic sealing with long-term stability [1]. The dictionary definition of the term hermetic is a seal that is completely gas tight or impermeable to gas flow. With MEMS technologies it means an air tight seal which will keep gases and moisture out of the package; hence, failure of the device due to condensing water vapor inside the package is avoided. The package samples, such as OLED and MEMS, are various, complicated and not standardized [2-4]. They proposed a severe challenge to packaging technology. The laser bonding technology has the advantage of local heating, small affected zone, fast heating and high efficiency. It has been researched in high air tightness package samples [5-6]. The key problem of laser package is the residual stresses and the reliability failure. In many cases, high temperature can cause the internal functional material of devices damage or failure. In 2005, Corning put forward a technique that uses low melting point glass material for glass-glass laser bonding, extending its field from MEMS to photoelectric [7].

With the development of modern microelectronics technology and optoelectronic technology, as well as the higher requirement of sealing product and environmental protection, sealing materials present a trend of low melting point, utilizing, high air tightness and sealing strength. Among them, the bismuth glass system has attracted great interests in industry, due to its possibility in replacing leaded glass [9]. American [10], South Korea [12] and Japan [11] have put forward some related patents in this field, but there are still imperfections.

In this paper, we base on low melting bismuth glass material as the research object, analyses the effect of CuO additive on laser absorption performance and the bonding quality. In this way, we achieved glass to glass bonding at a low laser power.

II. EXPERIMENT

Transmission laser bonding of glass to glass is carried out in four main steps: the preparation of glass paste, the deposition of the glass paste, the conditioning of the paste and finally complete the laser bonding.

The glass substrate used in this experiment is the corning Code 1737 glass. The glass frit contains two components: low melting point glass powder and additives. The glass powder adopted here is a kind of bismuth glass powder. Its melting point is about 500℃, and its average particle diameter is 5μm. The additive here is CuO. The wavelength absorption peak of CuO in glass is 800-1100nm. Adding CuO could improve the absorption of glass material for 810nm laser.

Control experiment is divided into four groups, the proportion of CuO in each group is respectively: 5%, 10%, 15%, 20%. Through the preparation of glass paste, the deposition of the glass paste and the conditioning of the paste, the glass frit is formed squarely on the glass substrate in 0.8mm wide and 15 microns thick. The process of the presintering heating temperature curve is shown in Fig. 1. Firstly we vapor the water around 130℃. In the second heat preservation area of 300℃, we burned out the organic matter. In the last stage of preservation, the glass frit melts and cures on the substrate.

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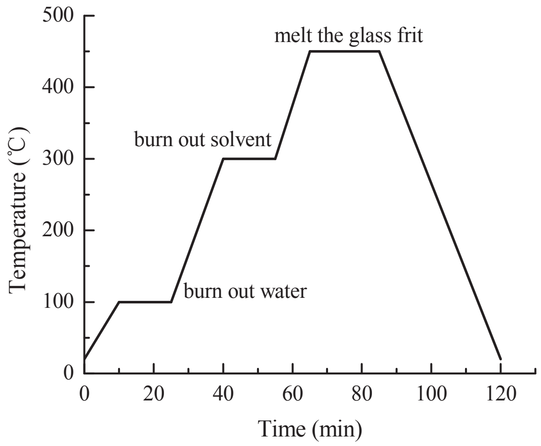


Fig. 1. Temperature history of the thermal conditioning step used in glass to glass bonding

The presintering process melts the glass frit and makes it solidification on the plate glass substrate to encapsulate the sample preparation into experiment. In the laser bonding process, align the presintered substrate and the other glass substrate, use the laser bonding equipment for encapsulation. The scanning heating process is shown in Fig. 2. The glass frit melts and wets adequately, and after cooling it forms a high intensity sealing. The laser used in the bonding process is 810nm, the power is 20w, the laser beam diameter is 1.0mm, and the scanning speed is 2.4mm/s.

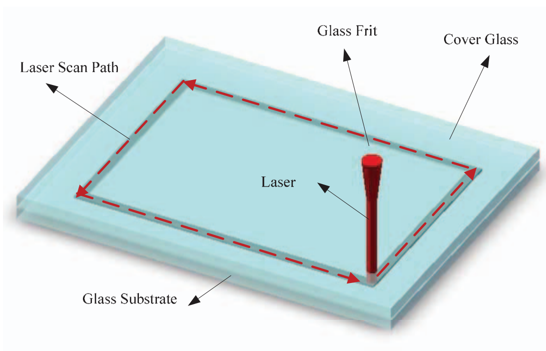
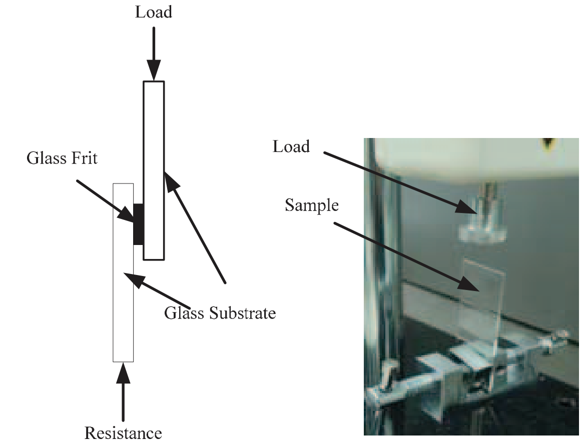


Fig. 2. Schematic of the laser bonding sample

In order to analyze the sealing intensity of laser bonding, this paper adopted the shear tester. The experimental sample of bonding area is designed to 0.8 * 5.0 mm^2, the shear force tester used here is HG-500. The specific experimental method is shown in Fig. 3.



(a) (b)

Fig. 3. (a) Schematic of shear force test, (b) Testing machine and sample

In order to deeply study the effect of CuO content on quality of packaging. This paper used the SEM and EDX analysis on the bonding interface morphology and element distribution of samples.

III. RESULT AND DISCUSSION

A. Sealing Surface and Cross-section Morphology

Fig. 4 is the sealing sample morphology of CuO content respectively 10wt%, 15wt%, 20wt%. Because of the 5wt% sample failed to bond, the result was not shown here. As the CuO content is 10wt%, after laser bonding cooling to the room temperature, tiny cracks appears on the glass substrate. As the CuO content is 15wt%, there is no obvious crack on the glass substrate. As the content of CuO is increased to 20wt%, the bonding interface of glass substrate appears cracks around the corner. Glass material layer absorption laser produced high temperature melt glass frit. If the absorption is too low, the temperature that the laser caused can’t fully melt the glass frit, and the melting frit doesn’t flow well. When it cools, the frit between the substrates bond loosely and cracks on the glass layer. Appropriate laser absorption rate makes the frit melts fully and forms a good bonding in the cooling process. But if the bonding temperature is too high, not only the frit but also the glass substrate will deform. When it cools to the room temperature, the glass substrates will crack around the corner on the frit paste because of the mismatching of thermal expansion coefficient (CTE) between glass frit and glass substrate.

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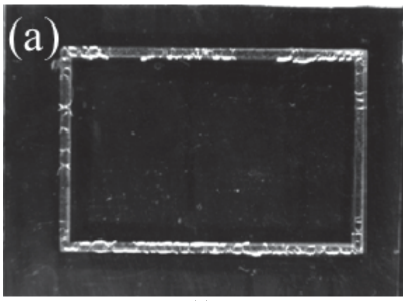

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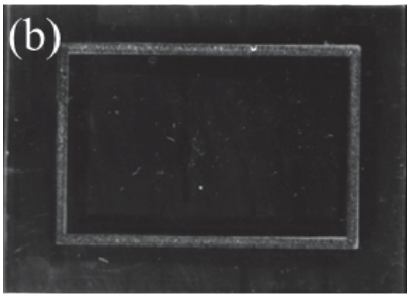

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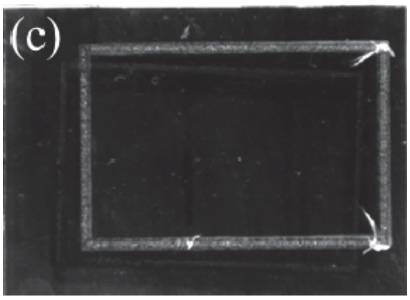
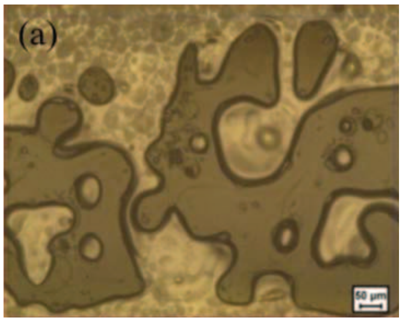


(c)

Fig. 4. Topography of laser bonding sample (a) 10wt%, (b) 15wt%, (c) 20wt%

The internal morphology of glass frit is shown in Fig. 5. Fig. 5a is for the 10wt% samples. The size of the sample is 30*40mm2. In this figure the frit only fills part of the bonded area, glass frit not fully melt to flow to cover the entire glass material area. In Fig. 5b and 5c, glass material melts and flow across the frit printed area. There are no holes generated through the glass material area. When the laser irradiates the glass frit, the residual gads can't be ruled out, and leaving mutually independent and small bubbles.



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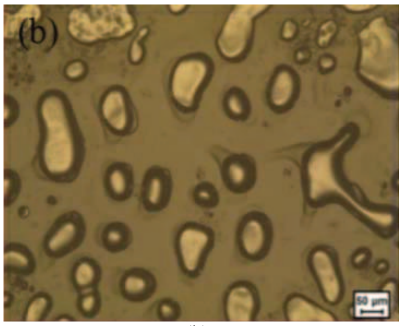

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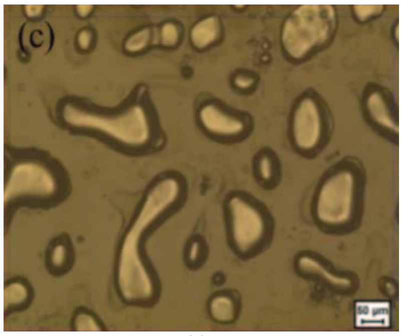

(c)

Fig. 5. The optical micrograph inside of the internal bonding section, the content of CuO (a) 10wt%, (b) 15wt%, (c) 20%

For further analysis of the properties of the bonding interface, using the SEM observes the microstructure of the bonding interface, the scanning result is shown in Fig. 6. The Fig. 6a is the sample of 10wt%, some not-molten glass particles and big bubbles appears. The interface between glass frit and substrates is smooth, and the thickness is 15 micron. The Fig. 6c is the sample of 20wt%, the glass frit and substrate bond closely. But the interface between glass frit and substrates bends obviously, the minimum thickness is 5 micron and the maximum is 15micron. The differences of thickness is mainly due to the percentage that frit melts and liquidity caused by molten glass.

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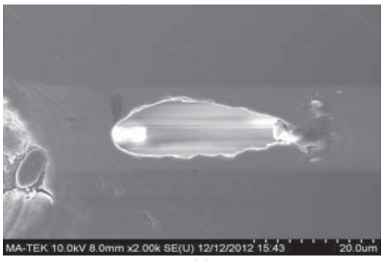

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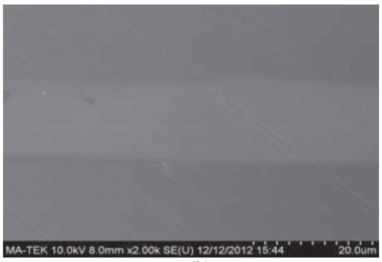

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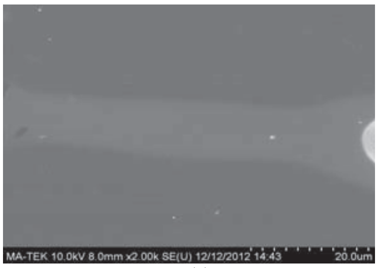


(c)

Fig. 6. The microstructure of the bonding section,the content of CuO (a) 10wt%, (b) 15wt%, (c) 20%

In the laser bonding process these four groups must use the same process parameter. Contract Fig. 4, 5 and 6, we can know that, the improvement of copper oxide content can effectively improve the laser absorption properties of glass frit. When the absorption of laser energy is insufficient, the frit is not fully molten, leading to the residual stress and cracks. If the absorbed energy is too high, the glass substrate may melts and deforms because of the high temperature. Only when adding suitable content of copper oxide could achieve a better bonding result.

B. *Shear Force Performance*

According to the American military standard MIL-STD-883G, a stress more than 25.0N on area of $0.8*5.0mm^2$ could meet the requirement of sample's bonding shear force. Samples of 5wt% laser bonding fails, we can't test the shear force. Figure 7 is the shear test result of three kinds of CuO content samples. Each data is the average shear force value of three groups of samples. The shear force of sample 10wt% is 81.0 N, 15wt% samples is about 125.0N, and the shear force of 20wt% samples is 113.7 N. They all satisfy the requirements of laser package on the properties of mechanics.

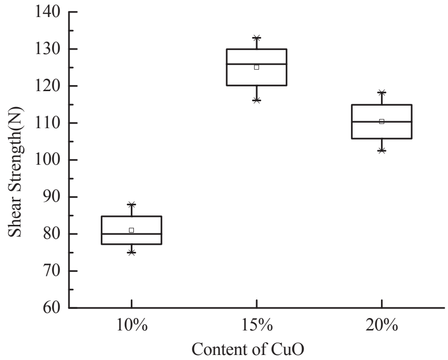
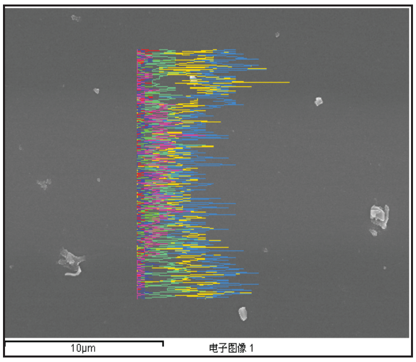


Fig. 7. Effect of CuO content on shear strength

In the analysis of rupture cross section, we found that the rupture section of 10wt% sample fracture directly from the adhesive layer, it is due to the low temperature produced by laser failing to fully melting glass frit. 15wt% and 20wt% samples fracture along the interface of substrate and frit. But few of the 20wt% samples fracture along the cracks on the substrate. The shear strength of 15wt% is stronger than the 20wt% one. Because it deforms in high temperature and its shear strength declines.

C. *Cross Section Spectrometer Analysis*

The result of shear test and interface morphology analysis both showed that the best encapsulation appears when the content of CuO is 15wt%. We further analyzed its interface EDS spectrum diagram, the result is shown in Fig. 8, As shown in Fig. 8a, the color line is the EDS scanning path, length of 15 micron. The color graph on the scan line shows the element content map, among them pink is for the content of copper. As shown in Fig. 8b, the glass frit layer near the substrate has significant lower copper content. The copper element diffused into the substrate through interface. Due to corning glass substrate doesn't contain copper, the existence of the copper in the transition layer showed an effective encapsulation.



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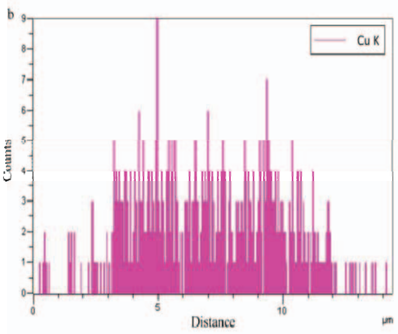

(b)

Fig. 8. (a) SEM of the bonding section, (b) Energy dispersive map of copper

IV. Conclusions

In this paper, we use CuO as additive into bismuth glass frit, studied its effect on properties of laser absorption and bonding quality of sample. Under the same laser bonding process parameters, we adjust the content of CuO, through SEM, shear test and EDS spectrum test, analysis the encapsulation quality. The study found that the adding CuO can effectively improve the laser absorption performance. When CuO content is too low, the temperature generated by absorbing laser cannot make frit fully melted, and the frit inside cohere loosely, some particle exists, cannot form effective bonding. If the CuO content of samples is too high, the frit and the substrate may deforms because of the high temperature. The contact interface between them will bend, and excessive internal stress is prone to crack. Additives and bonding process optimization can effectively achieve a better encapsulation.

V. Acknowledgments

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